Sefik EGRIBOZ (M.Sc., PMP, PSPO, ITIL)



SENIOR SOLUTIONS ARCHITECT

Enterprise Architecture | Enterprise Integration
Data Platforms | AI-Enabled Solutions

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PROFESSIONAL SUMMARY

Key results: **98%** fewer reported ATC callsign confusion events (**50 to 1** per season) with production AI | **99.8%** availability across **200+ integration points** | next-day to **near-real-time** data for **78 systems** | **EUR 8M** cumulative savings

Solution, integration and data architect, experienced in the IT of a large, transaction-intensive 24/7 airline group: designed, governed and delivered its integration backbone of **~50M annual transactions** on Kafka, IBM MQ and REST/SOAP APIs.

Data and AI end to end: an operational data store with canonical data model, master data management, ETL/ELT pipelines and data quality controls, and a graph neural network system in production for an **NP-hard** Quadratic Assignment Problem.

Sustained core operations through growth from 10M to 85M passengers a year as customer-side product owner for Lufthansa Systems NetLine, SchedConnect and flight planning across Turkish Airlines, AJet and SunExpress (a Lufthansa joint venture): a five-vendor RFP, POCs, due diligence and key-user training. Hands-on technical authority with **delivery accountability**.

RECENT PROFESSIONAL DEVELOPMENT

Career Break and Professional Development | **Istanbul** | **Sep 2024 – Present**

- Planned break for family and structured upskilling: **Harvard's CS50 AI** (12 Python projects: search, optimization, machine learning, neural networks, NLP), AWS Certified AI Practitioner and Cloud Practitioner, PSPO I and continued research on graph-based AI for safety-critical operations.

PROFESSIONAL EXPERIENCE

Principal Solutions Architect | **Turkish Technology** | **Istanbul, Turkiye** | **Jan 2023 – Sep 2024**

Official title: Distinguished Expert, the organization's highest technical grade.

- Cut reported ATC callsign confusion events by **98%**, from **50 to 1 per season** (sustained since), across up to ~1,500 daily peak-season flights, replacing earlier heuristic assignments with an in-house graph-based similarity and assignment optimization system aligned with EUROCONTROL CSST.
- Reframed callsign assignment as an **NP-hard** Quadratic Assignment Problem and built a scalable machine learning solution on **Graph Neural Networks (GNN)**.
- Implemented **Graph Attention Networks (GAT)** on Neo4j with PyTorch Geometric and feature engineering over phonetic, visual and semantic similarity.
- Reduced monthly incidents from **8 to 1** in 10 months as owner of roadmaps and technical direction for **six mission-critical operational systems** (1,000 users, ~100 concurrent, 5M annual transactions, EUR 1M budget).
- Sequenced change through **integration dependency management** and architectural constraints.
- Ran the NetLine/Ops replacement selection (**5 vendors, 3 shortlisted, 2 POCs**; a **EUR 1.5M** decision) and vendor evaluation of flight planning platforms (Jeppesen, CAE, Lido) through RFP/RFI, vendor management and business cases.
- Advised VP-level leadership on build-versus-buy, technology strategy and long-term R&D strategy.
- Held schedule mismatches at **zero** across two carriers and two reservations systems during AJet's migration from TROYA, the group's in-house reservations and inventory platform, to a separate third-party PSS.
- Ran SchedConnect-based synchronization, inter-carrier schedule distribution and schedule change management in a transition architecture.
- Established design authority (architecture review board) and design reviews bringing decisions under ISO 27001 compliance, software quality management and enterprise architecture governance, supporting regulatory audits and compliance.
- Led an **8-person cross-functional team** (project manager, 2 system analysts, 1 business analyst, 4 software engineers), with Scrum Master support as required, and rated team members' performance.
- Delivered **9 systems to production** as architecture and design decision owner.
- Defined the implementation approach, mentored engineers hands-on and cleared complex technical blockers.
- Shaped integration and technology strategy for next-generation operations platforms as **NetLine Subject Matter Expert** for Turkish Airlines, SunExpress and AJet, and as SME in callsign safety and operations control systems.

Senior Integration Architect | **Turkish Technology** | **Istanbul, Turkiye** | **Jan 2021 – Jan 2023**

Official title: Senior Expert, Integration Architect.

- Drove the enterprise integration transformation: standardized **200+ integration points across 86 systems** (**~50M annual transactions**) under one integration governance and data governance model.
- Set enterprise messaging and data streaming reference patterns, adopted group-wide for interoperability, maintainability, observability and operational reliability.
- Shortened new-integration lead time from **one week to one day**, cut integration-related incidents by **99%** and retired **47 legacy point-to-point interfaces** by reusing the group-wide reference patterns.
- Achieved **99.8% availability** (planned maintenance included) and zero reported unplanned downtime on mission-critical platforms serving **500+ concurrent users**, leading cross-functional delivery management through production deployment under high availability, disaster recovery design and SLA/SLO management.
- Eliminated dispatches into restricted airspace (**18** in the 12 months before go-live, **zero** since) by engineering a real-time early warning system across NetLine and FPM.
- Covered up to **~1,500 daily** peak-season dispatch legs with **72** automated checks and alarms on planned routes.
- Enabled **1.5M** affected customers a year to rebook online, even those already en route, by extending core-platform integration with the in-house reservations platform TROYA to real-time exception and disruption management.
- Propagated operations control decisions to reservation, inventory and customer-facing systems as they happened.
- Modernized the integration landscape onto an event-driven application and data integration platform on Apache Kafka (topic design, schema registry), IBM MQ and REST/SOAP APIs, sustaining **~50M annual transactions**.
- Applied microservices, enterprise integration patterns, API-first design and API lifecycle management, including naming conventions, API versioning and backward compatibility.
- Engineered the reliability semantics behind the estate's **99.8%** availability: publish/subscribe and asynchronous processing, idempotency in consumers, retries and exponential backoff, timeouts and error recovery.
- Built in dead-letter handling and poison-message processing, message ordering and durability, and replay of persisted streams for recovery.
- Owned the solution architecture artefacts that kept decisions traceable to a maintained target state architecture: solution blueprints, architecture roadmaps, High-Level and Low-Level Design (HLD/LLD), Architecture Decision Records (ADR) and non-functional requirements (NFR).
- Integrated a partner service behind an API gateway, handling API-key acquisition and token renewal on expiry.
- Raised release frequency from **1 to 4 a month** and **halved failed releases** by setting the estate's engineering practice: CI/CD architecture and DevOps release automation.
- Gated releases with GitOps configuration control, automated testing strategies, and performance and security testing.
- Ran **Scrum**-based agile delivery in **Jira** and **Confluence**: user stories with acceptance criteria, backlog management, sprint planning and Definition of Ready/Done.
- Caught **99%** of integration failures before users reported them and reduced mean time to detect from **one hour to one minute** with a real-time observability tool built across all live enterprise integrations.
- Based the tool on structured logging, correlation IDs and distributed tracing, and directed delivery of Grafana dashboards, metrics and alerting to support proactive issue detection.

Lead Solutions Architect and Platform Owner | **Turkish Airlines** | **Istanbul, Turkiye** | **Jun 2014 – Jan 2021**

Official title: Project Manager, Flight Operations.

- Designed and built a **Tier-0** (24/7 mission-critical) operational data store and evolved the operational plan data store (live since 2010) into a second Tier-0 platform.
- Took plan and operational-change delivery from **next-day to near-real-time** for **78 consuming systems**, replacing **70 separate data interfaces** by routing every change through both data stores, 24/7.
- Architected both as integration layers in front of Lufthansa Systems NetLine and the flight planning system, and as the authoritative **single source of truth** for enterprise reporting and downstream processing.
- Owned enterprise-wide operational data distribution for planning, operational and post-operation periods as the single integration point for **eleven downstream domains**.
- Fed planning, workforce, load planning, reservation and customer-facing systems, cargo, maintenance, DWH, revenue and cost accounting, SAP finance, HR/payroll and regulatory reporting.
- Processed **~100K update records a day** on average in peak season, 24/7 and near-real-time.
- Consolidated, transformed and enriched every update across seasonal plans through source-to-target mappings, ETL/ELT pipelines and integration rules under data quality and metadata controls.
- Architected batch, near-real-time and real-time distribution patterns that absorbed **~22.5M** data changes a year (~450K annual flights, ~50 changes per flight) before fan-out to large-scale distributed data processing and analytics.
- Set consumer-specific horizons, from a **rolling three-day window** to a **90-day window**.
- Fed operational changes (delays, cancellations) to customer-notification systems, triggering **75M notifications a year**, and operational data to loyalty and disruption hotel systems, so all three ran on the same data as operations.

- Designed operational data and integration architecture on the ODS pattern, standardizing core operational and plan data enterprise-wide: logical data models, entities and relationships, canonical data structures, ESB pattern applied twice.
- Evolved both Tier-0 platforms for **ten years** (2014 – Sep 2024) as de facto product owner with final authority over feature prioritization and architectural evolution.
- Coordinated operational stakeholders and cross-team dependencies for two Tier-0 platforms, carrying releases through production under ITIL-aligned **Site Reliability Engineering (SRE)** practices and continuous monitoring.
- Drove technical direction, integration and platform modernization for operational systems with a **EUR 6.5M** total programme budget (planning, scheduling, operations control) as lead architect.
- Worked end to end: system and business analysis, software architecture and hands-on coding of critical components through release, operational handover and 24/7 mission-critical production under ITIL.
- Managed NetLine's master and reference data (NetLine/Base) with the team: built and maintained the tables where users edit it and trained key users in master data management (MDM), keeping master data owned at its source.
- Built a reporting layer into both platforms on the same enriched data and fed it back into NetLine and the flight planning system, so enterprise reporting and operations shared one version of the data.
- Kept **850 NetLine users** working through the pandemic as NetLine product owner, coordinating the VMware solution.

Integration Solution Design Lead | Turkish Airlines | Istanbul, Turkiye | Jan 2007 – Jun 2014

Official title: Acting Team Leader.

- Delivered **EUR 8M** in cumulative savings from 2012 to 2023 (**EUR 800K** in the first year) with a multi-currency cost reconciliation engine built on a data pipeline of entry/exit timestamps and segment distances.
- Led a **6-person team** through the lifecycle of **100+ integrations** (requirements and system analysis, design, development, testing, documentation), harmonizing project plans and tracking cross-functional milestones per project.
- Rated team members' performance and interviewed technical candidates as Acting Team Leader; hiring decided by others.
- Architected the message integration and data mapping from core planning data to the in-house reservations platform (TROYA) under industry messaging standards, keeping inventory and reservations in step with cancellations, re-timings, equipment and routing changes, reinstated services and reason codes.
- Introduced SchedConnect (2007) and operated it as a standards-based B2B data exchange platform for recurring schedule data interchange with **54 partner carriers**.
- Headed solution design and delivery for operations, workforce and planning integrations across global technology vendors, building multithreaded real-time integration frameworks (2010) and contributing to an in-house Enterprise Service Bus (ESB) on Service-Oriented Architecture (SOA).
- Built a plan data store in 2010 to cut mainframe dependency: plans from a new flight planning system landed in Oracle and the data warehouse, easing SAP integration.
- Gave the fuel efficiency team plan data it rated far more accurate than the earlier Jeppesen feed.
- Developed the first heuristic callsign similarity and assignment system (phonetic, visual and semantic similarity), the groundwork later rebuilt as the graph-based AI solution once reclassified as NP-hard.
- Commended in writing in 2010 by Turkish Airlines' **Chief Pilot and Head of Flight Operations** for flight operations projects, the callsign project first among them, and personally by the **General Manager and CEO**.

Software and Integration Engineer | Turkish Airlines | Istanbul, Turkiye | Apr 2000 – Jan 2007

Official title: Analyst Programmer.

- Contributed to the first production implementation of the Lufthansa Systems NetLine suite.
- Cut costs by **EUR 100K** a year by designing and building the timetable publication system from scratch (Java, C, PL/SQL), so the scheduling department began preparing the timetable books itself.
- Turned worldwide schedules and partner flights into a PDF timetable for web download and print.
- Administered NetLine at application level with 24/7 availability responsibility (upgrades and patches on AIX, database schema upgrades) and built its bi-directional real-time interfaces to the IBM mainframe on IBM MQ.
- Drove the technical work of the first major legacy modernization from day one as an implementation team member, moving scheduling and operations control off the IBM mainframe (ADABAS/Natural) to Oracle on Unix (AIX) and NetLine.
- Designed and built a 24/7 real-time world map tracing every company flight in the air with simulated positions: architecture decisions, mainframe integration over IBM MQ and UDP multicast (Java, Oracle, ILOG JViews Maps).
- Ran **statistical analysis** of historical data to determine the methodology for the project that calculated average passenger and baggage weights; the methodology was adopted for weight-and-balance calculations.

CORE COMPETENCIES & TECHNICAL SKILLS (hands-on)

Integration Architecture and Schema Governance: Event-Driven Architecture, API Architecture, reference architectures, Apache Kafka with schema registry, schema versioning and backward-compatibility rules, controlled change across event-driven integrations, IBM MQ, Enterprise Service Bus (ESB), REST/SOAP APIs; GraphQL (design level).

Data Architecture and Governance: Enterprise Data Architecture, Data Modeling, Canonical Data Model, Operational Data Store (ODS), Data Integration Architecture, Data Pipeline Architecture, master data management (MDM), data lineage through source-to-target mappings, data quality and metadata controls.

Data and AI Platform Architecture: Graph data models (Neo4j), GNN-based assignment and decision support (PyTorch, PyTorch Geometric), data pipelines, MLOps, data warehouse and BI feeds, real-time analytics.

Engineering Platforms & Languages: Oracle DB, Docker and containerization, Linux (RHEL) and Unix administration (AIX, HP-UX), application administration on Windows Server; Languages: Python, Java, C, PL/SQL, shell scripting (Bash, KSH), YAML and JSON; Git and Jenkins; UML; multithreaded real-time and database-driven systems; 50+ applications and integrations to DWH, SAP and MRP, with incident analysis and performance tuning.

Executive Leadership and Delivery: Digital transformation, transformation governance, executive (C-level) stakeholder management, PMI/PMP delivery governance, operational readiness, customer consulting and workshop facilitation.

EDUCATION

• Ph.D. (coursework)	Business Administration (Finance)	Yeditepe University	Istanbul
• M.Sc.	Management Engineering; thesis on data warehousing for mass customization	Istanbul Technical University	Istanbul
• B.Sc.	Computer Science Engineering	Istanbul University	Istanbul

LANGUAGES

• English	C1	Full Professional
• German	A2	Studying towards B1
• Turkish		Native

CERTIFICATIONS

• Project Management Professional (PMP), Credential ID 1225730	PMI	2008
• Professional Scrum Product Owner I (PSPO I), Credential ID 1347372	Scrum.org	2026
• ITIL v3 Foundation, Candidate No. 990453483	BCS, The Chartered Institute for IT	2013
• AWS Certified Cloud Practitioner (CLF-C02)	Amazon Web Services (AWS)	2026
• AWS Certified AI Practitioner (AIF-C01)	Amazon Web Services (AWS)	2026
• CS50's Introduction to Artificial Intelligence with Python (12 projects)	Harvard University (CS50)	2026
• Manager Ready (DDI) leadership simulation: exceptional rating	TAG Taurus Group	2024
• AI, Cloud and Container Foundations: OCI AI Foundations; Agentic AI Foundations; IBM SkillsBuild; PMI AI (CPMAI, GenAI); Docker Foundations Professional Certificate	Oracle, IBM, PMI, Docker	2026
• Lufthansa Systems SchedConnect vendor training (two weeks)	Lufthansa Systems	2007